

laying the groundwork for future autonomous deep-space missions.

The James Webb space telescope

The James Webb Space Telescope, operational since mid-2022, functions at the Sun-Earth L2 point, 1.5 million kilometres from Earth, and requires extremely precise pointing to capture faint infrared signals.

Its attitude control system fuses star trackers, fine guidance sensors (FGS), and gyroscopes via a radiation-hardened PowerPC computer that runs adaptive algorithms. The FGS uses a mercury-cadmium-telluride infrared detector with cryogenic amplifiers operating near 40K to minimise Johnson noise, enabling photon detection and milliarcsecond centroiding in firmware.

Onboard fault detection resolves conflicts, such as stray light or cosmic rays, by adjusting Kalman filter covariances to downweight unreliable data. Hemispherical resonator gyroscopes detect Coriolis shifts in fused-silica hemispheres, while analogue-to-digital closed-loop control maintains short-term stability. The fusion algorithm propagates attitude using gyro data between star-tracker updates, which corrects drift and preserves absolute accuracy.

Perseverance Mars helicopter Ingenuity

The Ingenuity helicopter exemplifies extreme autonomous navigation under severe resource constraints, operating its entire navigation suite on just fifteen watts.

The system integrates a downward-facing camera, a commercial automotive-grade inertial measurement unit, and a laser time-of-flight altimeter within an efficient architecture. A Qualcomm Snapdragon 801 mobile processor demonstrates the viability of commercial electronics in space applications while executing visual odometry algorithms that track surface features between frames to compute velocity and position.

The design exploits the processor's integrated digital signal processor and graphics processing unit to accelerate computationally intensive correlation tasks. The inertial measurement unit supplies gyroscope and accelerometer data at 500Hz via serial peripheral interface buses. A loosely coupled extended Kalman filter, implemented in C++ on ARM cores, fuses visual odometry with inertial integration.

This adaptivity proves essential. When dust devils or lighting variations corrupt visual measurements, the filter autonomously rejects outliers and transitions to inertial dead reckoning until visual measurements are reliable again.

Recent technological advances

Several trends enable robust autonomous navigation beyond labs.

- **First**, cheaper, smaller, and more capable sensors, including compact star trackers, low-SWaP lidars, high-dynamic-range cameras, and photonic integrated-circuit gyroscopes, expand the options for small- and medium-sized missions. European Horizon-funded ultra-low-power photonic gyros and specialised lidars promise higher precision at lower mass and power.

- **Second**, radiation-tolerant FPGAs and space-qualified AI accelerators support real-time inference of compact neural networks for feature extraction and anomaly detection within strict power budgets, while reconfigurable logic enables hardware-in-the-loop verification and safe rollback.

- **Third**, advances in algorithmic maturity in heterogeneous data fusion and explainable machine learning have introduced adaptive online covariance estimation, efficient particle filters, federated architectures, and hybrid physical-data-driven models that balance safety, traceability, and flexibility, and have been validated through theory, simulation, and on-orbit testing.

- **Finally**, stronger systems

engineering practices, autonomy standards, formal verification of critical paths, and realistic fault simulation tools simplify certification and reduce programme risk.

Emerging technologies: X-ray pulsar navigation

X-ray pulsar navigation leverages the universe itself as a natural beacon system. Millisecond pulsars—rapidly rotating neutron stars emitting highly periodic X-ray pulses—function analogously to celestial atomic clocks. By detecting these signals and comparing pulse arrival times with predicted schedules, spacecraft can determine their position with an accuracy of hundreds of metres over interplanetary distances.

Electronic implementations typically use silicon drift detectors or scintillator-photomultiplier combinations. Silicon drift detectors generate a lateral electric field that guides photoelectrons toward a small anode, thereby improving charge collection and energy resolution. The analogue front end performs charge-sensitive preamplification, shaping amplification to optimise signal-to-noise ratio, and baseline restoration to prevent pile-up at high count rates. The digital back-end performs pulse-height analysis, analogue-to-digital conversion, and photon time tagging with nanosecond precision.

The Station Explorer for X-ray Timing and Navigation Technology demonstration aboard the International Space Station, conducted between 2017 and 2018, validated this concept using neutron-star timing data. The system achieved autonomous position determination within approximately 10km, a notable result for a purely passive measurement approach.

Intelligent adaptive algorithms

We observe that integrating machine learning, particularly neural networks, into spacecraft navigation represents a shift in how adaptive fusion operates. Traditional methods